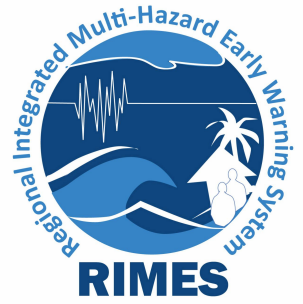




IMD Rainfall Information System (IRAINS)

Hydromet Division – Mausam Bhawan

1. Introduction

IRAINS (IMD RAinfall INformation System) is a specialized platform developed for the Hydromet Division, by IRU (IMD RIMES UNIT) using cutting edge technologies by focusing on the efficient tracking and analysis of rainfall data across various regions in India. This application serves as a central hub for rainfall data collected from multiple meteorological stations and is intended to support operations within the Hydromet Division. IRAINS offers a wide range of features that empower HQ, MC, and RMC users to monitor daily rainfall, examine historical data, and generate graphical and tabular reports, fostering transparency and accuracy in meteorological data handling. The system supports real-time data collection, statistical analysis, and customizable data entry, designed to meet IMD's mission of providing precise, accessible, and actionable climate information to its users.

IRAINS is built with scalability in mind, handling data at multiple levels, from district and state to national coverage. It integrates interactive maps, extensive filtering options, and dissemination capabilities, enhancing data visibility and fostering timely responses to weather-related events. With an intuitive interface and comprehensive data analysis features, IRAINS is instrumental in addressing critical meteorological needs.

2. Overview

IRAINS is a web-based system designed to streamline rainfall data management for the Hydromet Division, Indian Meteorological Department (IMD). This platform employs an interactive, map-centric interface powered by Leaflet, allowing users to visualize rainfall data across districts, states, subdivisions, and regions with color-coded indicators that display actual, normal, and departure values. IRAINS also supports detailed station-level data entry, verification, and analysis, enhancing data accuracy and accessibility. IRAINS is designed to support the following functionalities:

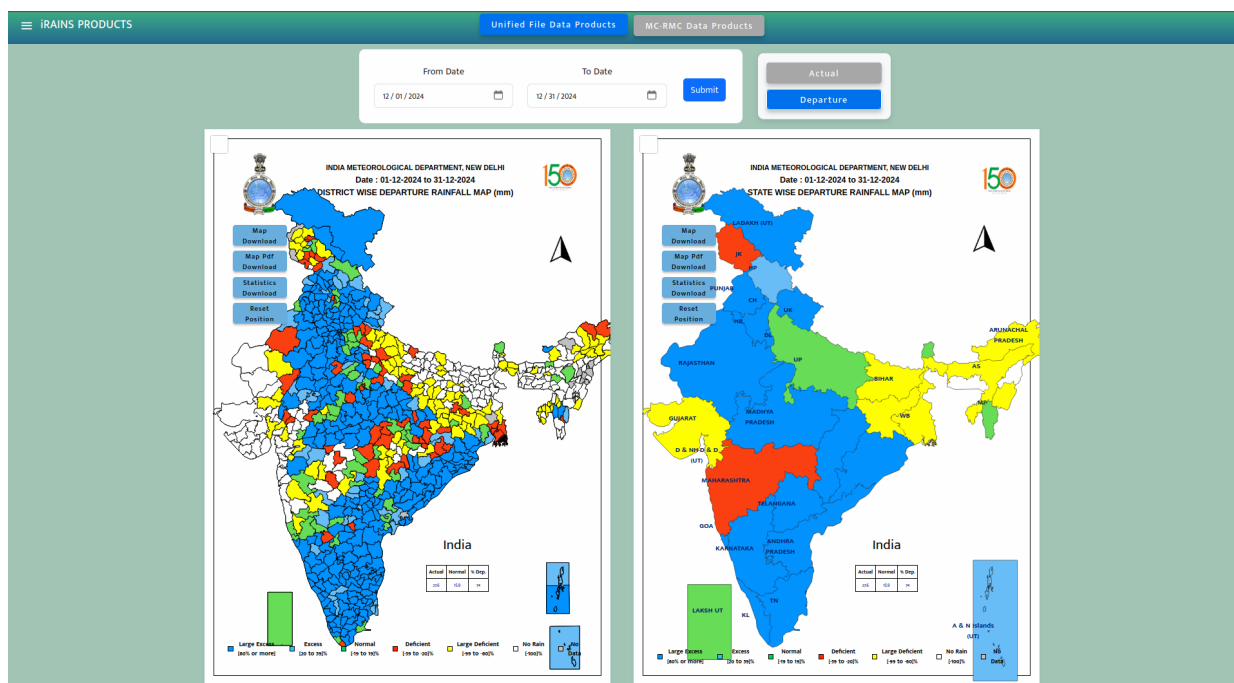
- **Data Visualization:** Interactive maps provide a quick and intuitive view of rainfall data, displaying Actual, Normal, and Departure values for specific periods. This functionality enables users to assess rainfall trends at a glance.
- **Data Input and Verification:** MC and RMC users contribute station-level data, while HQ users have the authority to verify and oversee the data, ensuring completeness and reliability.
- **Analysis and Reporting:** Graphs, statistics, and tables within IRAINS allow for in-depth analysis, with options to download and share insights, aiding in forecasting and historical data analysis.
- **Automated Notifications:** IRAINS automates email notifications, sending reminders to respective MC and RMC offices about stations with pending updates, as well as verification reminders with lists of stations requiring review.
- **Station Mapping and Comparison:** Users can view each station's details on a map, locate nearby stations for comparison, and view a 30-day running rainfall graph for each station.
- **Dissemination:** IRAINS facilitates data communication through emails with attachments, supporting organized and logged dissemination of critical rainfall information.
- **Significant Rainfall:** Enables users to define two threshold values to filter and view data within a specified range, providing detailed insights into significant rainfall events.
- **Yearly Station Statistics:** This section allows MC and RMC users to access rainfall data in a unified file format, filtered according to their operational needs. Designed for convenient data aggregation, it delivers the required data for comprehensive reports or analysis.
- Additionally, **flood maps are still under development**, and the **station statistics feature is in progress**.

IRAINS is a vital tool for the division, offering centralized data access and localized data management through differentiated access levels for HQ, MC/RMC, and Guest (Public) users. This structure ensures data integrity across all user roles, fostering effective decision-making and seamless coordination within the Hydromet Division.

3. Features and Key Pages

Home Page

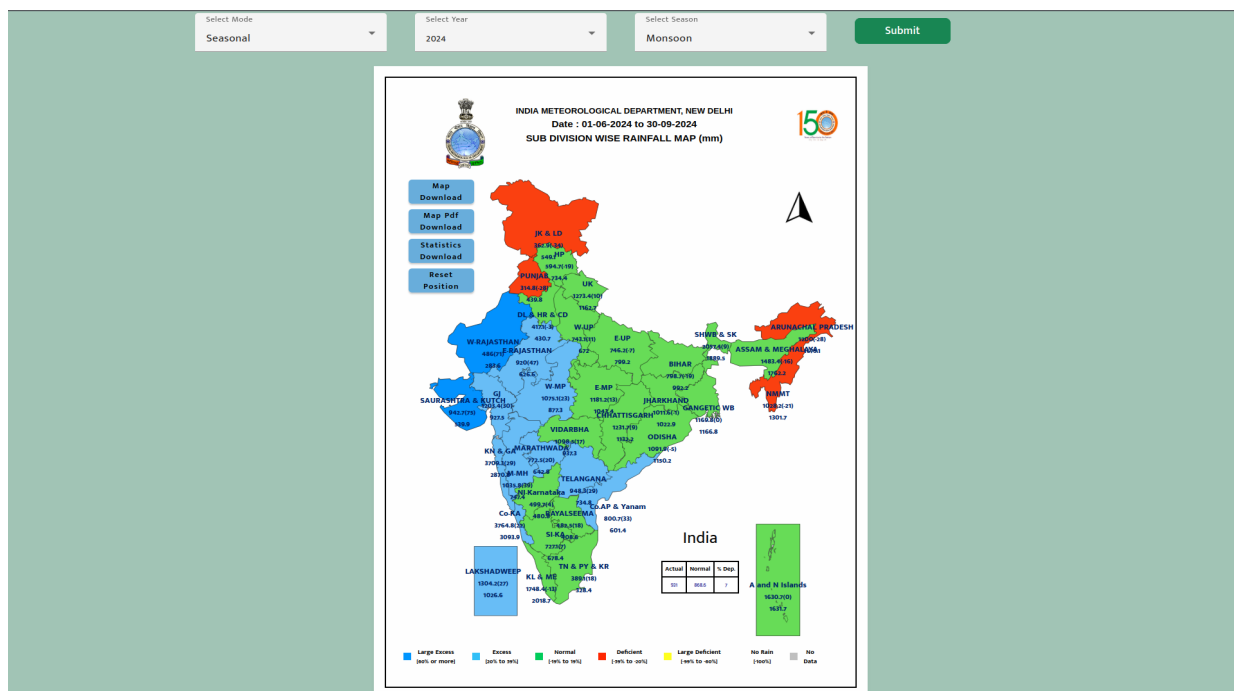
- **Unified File Product:** The Unified File Data Products, which compile MK rainfall data, serve as a temporary solution until MC and RMC centers fully transition to direct data entry within the system. This feature impacts the scope and format of the rainfall data presented.
- **MC-RMC Data Product:**
This section displays maps generated from the data entered by MCs and RMCs through the data entry page.
- **Date Range Selection:** An input section enables users to specify the data range, refining the dataset to display rainfall data within a chosen timeframe.
- **Actual Button:** Sets the color coding according to the actual rainfall values, enabling users to visualize the actual rainfall map effectively.
- **Departure Button:** Adjusts the color coding based on departure values, allowing users to visualize the departure map with clarity.
- **Map Visualizations:** Four interactive maps (District, State, Subdivision, Region) display color-coded rainfall data, with values for actual, normal, and departure rainfall:
- **Interactivity:** Users can zoom, pan, reset and toggle full-screen views.
- **Data Downloads:** Each map includes options to download A4-sized maps with color coding, export as PDFs, and download tabular statistics in both PDF and Excel formats.
- **Navigation Bar:** Provides access to various rainfall data features, allowing users to easily navigate between rainfall maps, graphs, statistics, and more.



(fig 3.1 : Home Page)

Rainfall Map

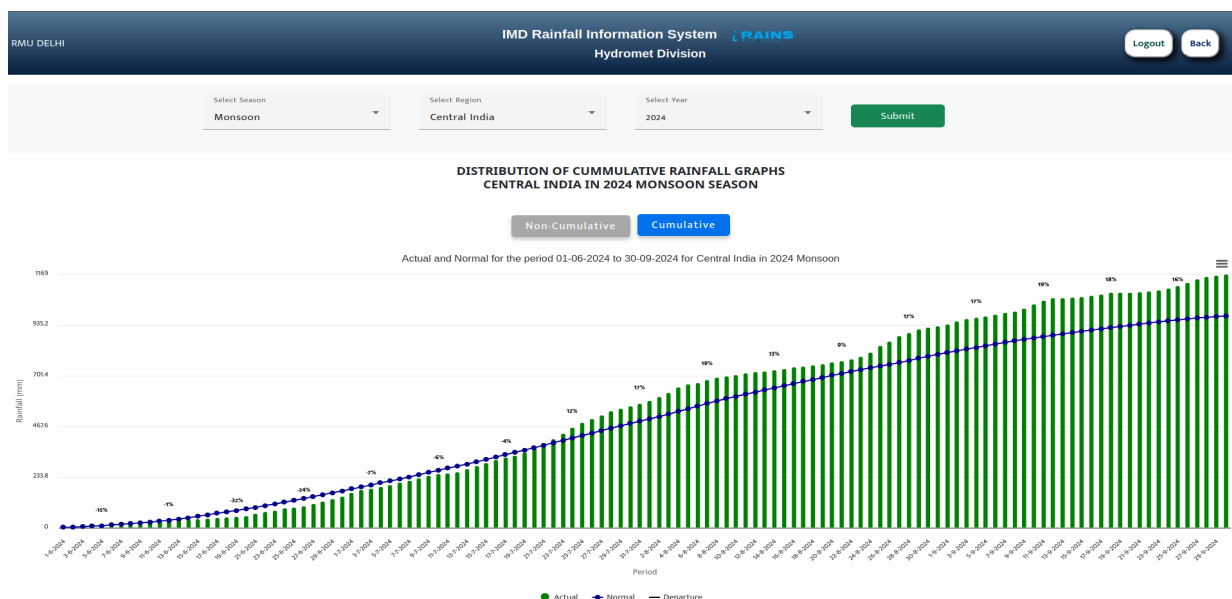
- **Daily, Weekly, Seasonal, Annual, and Cumulative Departure Maps:** Users can view actual, normal, and departure values at district, state, subdivision, region, and country levels. They can seamlessly switch between daily, weekly, seasonal, annual, and cumulative views to analyze rainfall data across various timeframes.
- **Normal Rainfall Maps:**
- **State-wise and District-wise Normal Rainfall:** Displays standardized normal rainfall data with intuitive color coding for easy interpretation.
- **Real-Time Rainfall:** Provides monthly, seasonal, and annual rainfall data, enabling immediate trend analysis through color-coded visualizations.
- **Download Features:** Every rainfall map can be downloaded, ensuring users have offline access to important data visualizations.



(fig 3.2 : Map representing normal and departure)

Rainfall Graphs

- **Graphical Analysis:** Provides a visual comparison of actual vs. normal rainfall, with the x-axis representing the date and the y-axis showing rainfall in millimeters. Users can analyze both cumulative and non-cumulative data to identify trends.
- **Inputs:** Fields for selecting the season (e.g., monsoon, winter), region, and year allow users to perform focused analyses.
- **Non-Cumulative Button:** Sets the graph view to non-cumulative, enabling users to observe fluctuations in actual and normal rainfall values over time.
- **Cumulative Button:** Adjusts the graph view to cumulative, providing a comprehensive visualization of cumulative rainfall trends for actual and normal values.
- **Export Options:** Graphs can be downloaded for offline use, making them ideal for presentations and in-depth analyses.



(fig 3.3 : normal and actual graph)

NWP Rainfall Product

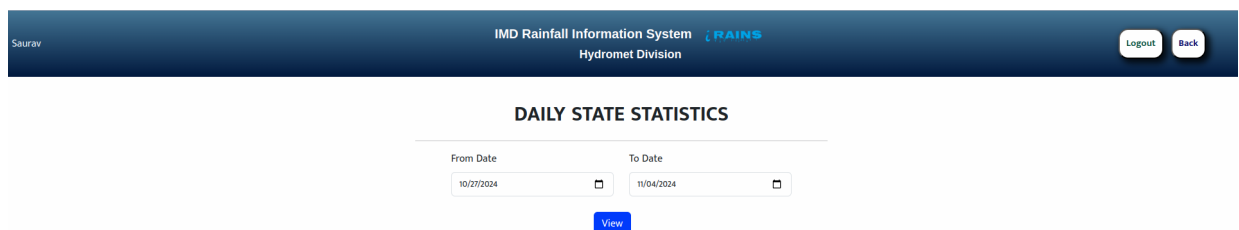
- **External Redirection:** Directly links to the NWP site, where users can access additional IMD rainfall products. This integration offers extended resources without disrupting the user's workflow in IRAINS.



(fig 3.4 : NWP Products)

Rainfall Statistics

- **Statistical Views:**
- **Daily and Weekly Cumulative Data:** Provides real-time statistics, enabling users to review data trends at the district, state, subdivision, region, and country levels.
- **Five-Year District-Wise Data:** Offers a historical overview of rainfall data over the past five years, facilitating in-depth district-level analysis.
- Supports downloadable formats for long-term data storage and offline reference.



(fig 3.5 : rainfall statistics)

Rainfall Departure

- **Departure Data Tables:** Displays departure values, allowing users to review variations in rainfall through a tabular format.
- **Input Filters:** Users can filter data by weekly/cumulative values, district/subdivision, season, and year, ensuring relevant data is easily accessible.
- **Export Functionality:** Users can export these values in a downloadable format.

The screenshot shows the 'RAINS' (Rainfall Information System) interface. At the top, it says 'IMD Rainfall Information System' and 'Hydromet Division'. There are filters for 'Select Mode' (Weekly), 'Select Map' (Subdivision), 'Select Season' (Monsoon), and 'Select Year' (2024). A 'Submit' button is present. Below the filters, a box displays 'Subdivision - Week By Week Departures (Weekly)' with a 'Download' button. The main table shows rainfall departure values for 18 subdivisions across 18 weeks. The values are color-coded: red for negative, yellow for zero, and blue for positive.

S.NO	Subdivision	05-06-2024	12-06-2024	19-06-2024	26-06-2024	03-07-2024	10-07-2024	17-07-2024	24-07-2024	31-07-2024	07-08-2024	14-08-2024	21-08-2024	28-08-2024	04-09-2024	11-09-2024	18-09-2024	25-09-2024	30-09-2024
1	CHHATTISGARH	60	-45	-75	-30	30	-45	-37	86	70	60	-34	-38	24	-47	131	-81	-31	151
2	EAST MADHYA PRADESH	-60	-66	-64	-29	28	-10	-23	45	26	89	-6	-24	52	-53	81	53	-61	150
3	GUJARAT REGION	-98	-32	-87	-16	39	-35	-15	31	53	5	-46	-71	393	93	62	-91	-44	425
4	KONKAN & GOA	-89	23	-45	13	15	75	89	109	31	14	-56	-81	145	-25	38	-16	26	191
5	MADHYA MAHARASHTRA	-52	123	28	-8	15	20	42	93	132	50	-55	34	248	12	-12	-81	12	78
6	MARATHWADA	-43	159	-7	-23	0	4	76	23	29	-62	-42	17	47	220	-30	-95	9	30
7	ODISHA	81	-60	-76	-32	21	-32	-30	56	-6	24	-48	-20	-2	-35	52	-4	7	87
8	SAURASHTRA & KUTCH	-99	-83	-49	-1	106	-5	-11	266	-32	-28	-14	-61	870	281	-38	-30	-98	424
9	VIDARBHA	3	19	-56	-16	24	23	-3	161	56	-8	58	-32	71	30	51	-99	23	199
10	WEST MADHYA PRADESH	14	38	-35	38	8	23	8	-6	54	57	-10	-34	76	-11	68	-71	-28	409
11	ARUNACHAL PRADESH	7	-66	-9	-29	71	10	-45	51	66	7	32	-73	39	-37	-62	-95	-72	15
12	ASSAM & MEGHALAYA	51	38	-11	-43	63	-26	-13	91	-63	19	23	2	-60	-53	-63	-86	-78	50
13	BIHAR	-39	-77	-75	-45	83	23	-34	-88	-84	73	-3	-32	-23	-71	-43	2	-70	252
14	GANGETIC WEST BENGAL	-32	-82	-80	-77	-14	33	27	31	0	132	-21	-23	116	-67	53	301	-41	89
15	JHARKHAND	4	89	-89	-63	23	-46	-34	-43	-5	153	0	-10	60	62	33	225	50	88
16	NMMT	-46	-50	7	-43	52	-34	-60	-43	-42	21	35	96	18	-2	-40	-66	-75	25
17	SHWB & SIKKIM	44	32	146	15	60	80	-2	-50	-54	42	-19	-62	-69	-56	-58	-48	-62	289
18	DELHI AND HARYANA AND CHANDIGARH	-78	-88	-97	-18	33	17	-75	-49	-53	32	71	7	26	41	71	104	-65	154

(fig 3.6 : Weekly Departure)

Info Upload

- **Data Upload:** Facilitates uploading of critical rainfall-related documents, including:
- **QPF Verification Report**
- **Design Storm Analysis**
- **Rainfall Report**

(fig
file

The screenshot shows a 'File Upload' form. It has a title 'File Upload' and a subtitle 'Select Section'. Below this, there is a text input field containing 'QPF VERIFICATION REPORT'. Underneath the input field is a 'Choose File' button with a paperclip icon. At the bottom of the form is a large blue 'Upload' button.

3.7 :

upload)

QPF Verification Report

- Overview:** Displays previously uploaded QPF verification reports, offering essential verification for rainfall predictions.

Design Storm Analysis

- Display:** Lists uploaded design storm reports, a valuable feature for assessing extreme weather events.

Rainfall Report

- Statistics Overview:** Lists uploaded rainfall statistics, which are essential for generating accurate weather forecasts.

Data Entry Page

- Detailed Entry System:** MC/RMC users can enter rainfall data at the station level, filtering entries by region, MC, RMC, state, district, and date. Users must enter data daily before 12:30 PM. If they fail to do so, an automated email is sent, listing the stations that need to be updated.
- Additional Tools:**
- Download:** Users can download the filtered station list, enter rainfall data into the provided Excel file, and upload it back to the site.
- Upload Station:** Enables bulk uploads of new stations using Excel, with sample templates provided for guidance.
- Upload Rainfall:** Facilitates bulk uploads of daily rainfall data by allowing users to pass an Excel file and update all stations at once.
- Individual Station Update:** Users can update data for individual stations directly by clicking on the rainfall value in the user interface.

(fig 3.8 : Entry)

CENTRAL INDIA

BHUBANESWAR

Select RMCs

ODISHA

ANGUL

11/05/2024

Submit

STATIONS DATA

Download

Choose File

No file chosen

Upload Stations

Sample File

Choose File

No file chosen

Upload Rainfall

Sample File

S.NO	STATION ID	STATION NAME	RAINFALL	TYPE	NEW/OLD	LAT° N Eg : 21.3443	LONG° E Eg : 85.3453	YEAR OF ACTIVATION	ACTION
1	1070701000002	ATHMALIK	-9999	AWS ORG ARG	OLD NEW	20.7167	84.5333	21-02-2024	Edit Delete
2	1070701000010	TALCHER	-9999	AWS ORG ARG	OLD NEW	20.9500	85.2500	21-02-2024	Edit Delete
3	1070701000003	BANARPAL ARG	-9999	AWS ORG ARG	OLD NEW	20.8333	85.2000	21-02-2024	Edit Delete
4	1070701000004	CHENDIPADA	-9999	AWS ORG ARG	OLD NEW	21.0000	84.8667	21-02-2024	Edit Delete
5	1070701000005	KANIHA ARG	-9999	AWS ORG ARG	OLD NEW	20.0833	85.0500	21-02-2024	Edit Delete
6	1070701000006	PALLAHARA	-9999	AWS ORG ARG	OLD NEW	21.4333	85.1833	21-02-2024	Edit Delete
7	1070701000008	RAJKISHORENAGAR	-9999	AWS ORG ARG	OLD NEW	20.9667	84.4667	21-02-2024	Edit Delete
8	1070701000011	TIKARPARA	-9999	AWS ORG ARG	OLD NEW	20.5000	83.3333	21-02-2024	Edit Delete
9	1070701000009	RENGALI	-9999	AWS ORG ARG	OLD NEW	21.2500	85.0167	21-02-2024	Edit Delete
10	1070701000001	ANGUL	-9999	AWS ORG ARG	OLD NEW	20.8333	85.1000	21-02-2024	Edit Delete

Add Station

Data

Significant Rainfall

- **Threshold Filtering (No Range):** Allows users to specify a threshold value to view data above a particular level, offering insights into significant rainfall events.
- **Threshold Filtering (Custom Range):** Enables users to define two threshold values to filter and view data within a specified range, providing detailed insights into significant rainfall events.
- **Data Download:** Filtered data can be downloaded for record-keeping and further analysis.

Select Rainfall Filter Type

Custom Range

No Range

Select Date

12 / 01 / 2024

Enter Rainfall Range

2

6

mm

Submit

Significant Rainfall Data

Download

Met Subdivision	District Name	Station Name	Rainfall (mm)
-----------------	---------------	--------------	---------------

(fig 3.9 : Significant Rainfall)

Verification

- **Data Verification:** MC users are responsible for verifying station data accuracy. The system provides filters for region, MC/RMC, state, district, and date to streamline and enhance the efficiency of data validation. Verification must be completed by **1:15 PM**. If not verified by the deadline, an automated email is sent to the respective MCs and RMCs, listing the stations requiring attention. After **1:30 PM**, the data is considered for map generation regardless of its verification status.

CENTRAL INDIA

BHOPAL

Select RMCs

MADHYA PRADESH

ANUPPUR

11/04/2024

Submit

STATIONS DATA

S.NO	STATION NAME	STATION ID	RAINFALL	Status	Verified Date Time
1	BIJURI	1020500100004	-999.9	VerifiedNot Verified	-
2	PUSHPRAJGARH	1020500100007	-999.9	VerifiedNot Verified	-
3	AMARKANTAK	1020500100001	-999.9	VerifiedNot Verified	-
4	VENKATNAGAR	1020500100008	-999.9	VerifiedNot Verified	-
5	BENIBARI	1020500100003	-999.9	VerifiedNot Verified	-
6	ANUPPUR-AWS	1020500100002	-999.9	VerifiedNot Verified	-
7	KOTMA	1020500100006	-999.9	VerifiedNot Verified	-
8	JAITHARI	1020500100005	-999.9	VerifiedNot Verified	-

Verify

(fig 3.10 : Data Verification)

Verification HQ

- **Comprehensive Data Verification:** HQ users can review MC/RMC data to assess station statuses, including updates, verifications, and pending records. The user interface displays a detailed list of MCs and RMCs, showing the total number of stations, updated stations, non-updated stations, verified stations, and non-verified stations for each MC and RMC.

11/05/2024

Submit

STATIONS DATA						
S.NO	MC or RMC	TOTAL STATIONS	UPDATES		VERIFICATIONS	
			UPDATED STATIONS	NOT UPDATED STATIONS	VERIFIED STATIONS	NOT VERIFIED STATIONS
1	MC RAIPUR	147	0	147	0	0
2	MC SHIMLA	115	0	115	0	0
3	MC HYDERABAD	351	0	351	0	0
4	MC BHOPAL	403	0	403	0	0
5	MC AHMEDABAD	366	0	366	0	0
6	RMC MUMBAI	289	0	289	0	0
7	MC BHUBANESWAR	317	0	317	0	0
8	RMC NAGPUR	143	0	143	0	0
9	RMC GUWAHATI	365	0	365	0	0
10	MC PATNA	268	0	268	0	0

(fig 3.11 : Data Entry)

Dissemination

- **Communication Management:** Send emails with attachments, create distribution groups, and maintain logs, supporting coordinated communication efforts.

Send Email

Email Group

Email Log

Send Email

Send To

Manual Email

Email

rncdelhi@gmail.com

Subject

Rainfall Alert

Message

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting,

Attachments

Choose File

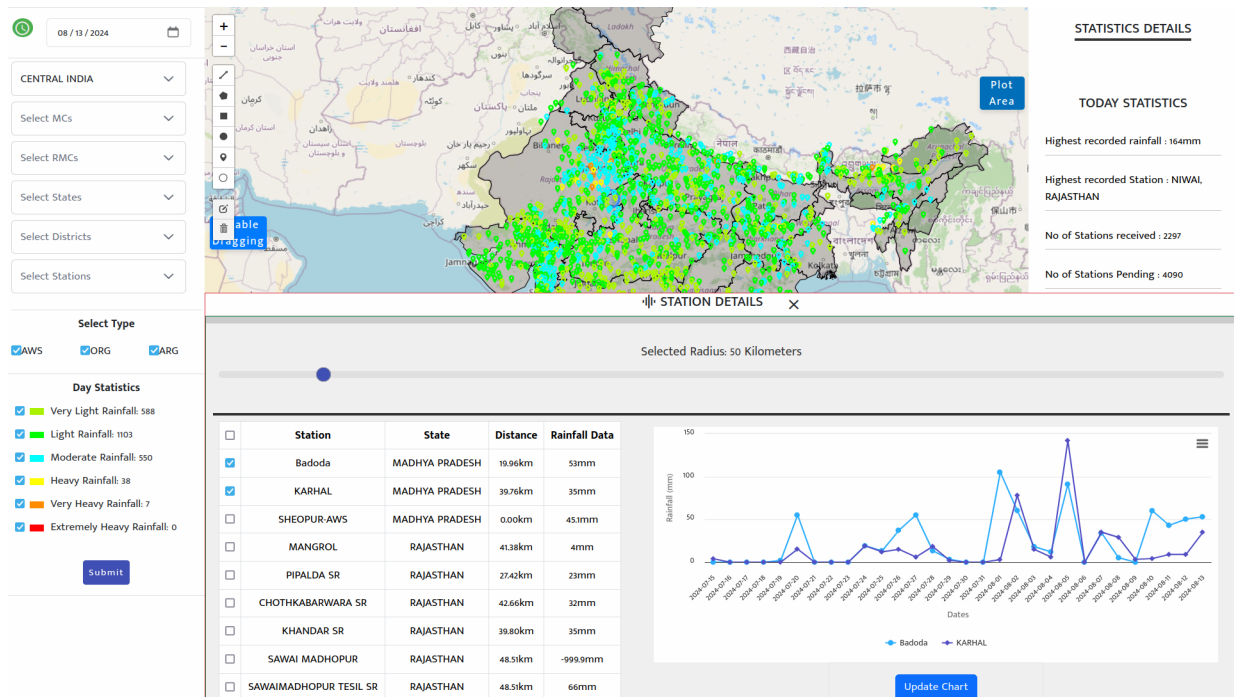
No file chosen

Send

(fig 3.12 : Send Email)

Station Statistics

- **Map-Based Insights:** Displays all station data with icons and color codes. Users can filter data by region, MC/RMC, state, district, and date.
- **Station Details:** Clicking on a station provides a 30-day running graph, showing rainfall records and other essential data.
- **Data Comparison:** Allows comparison of rainfall data across stations, with graphs available for selected stations.



(fig 3.13 : Station Statistics)

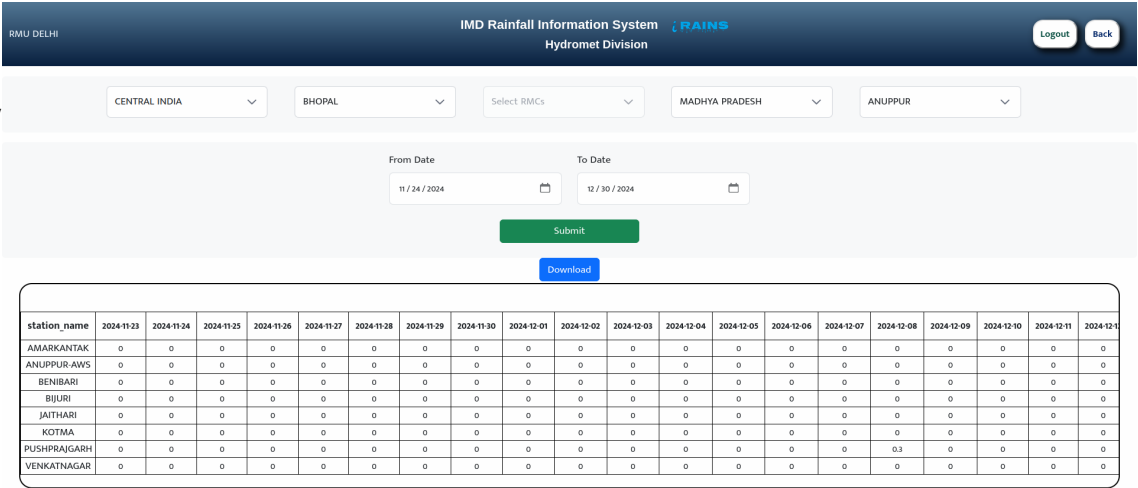
Yearly Station Statistics

- **Unified Data Retrieval:** This section enables MC and RMC users to access rainfall data in a unified file format, filtered according to their operational needs. Designed for convenient data aggregation, it provides the required data for comprehensive reports or analysis.
- **Filtering Options:**
 - **Region Selection:** Users have access to a broader "Region" filter, allowing for higher-level data aggregation across multiple centers.
 - **MC and RMC Selection:** Users can specify the Meteorological Center (MC) and Regional Meteorological Center (RMC) for targeted data.
 - **State and District Selection:** Enables filtering down to the state and district levels, ensuring localized data retrieval.
 - **Date Range:** The "To Date" and "From Date" fields define the date range, refining data selection to specific timeframes.

1. Usage and Importance:

- This feature supports MCs and RMCs in generating consolidated reports, tracking historical data, and planning for seasonal rainfall trends.
- Streamlines data access for quicker reporting and analysis during critical periods, such as monsoon season, aiding timely decision-making across regions.

(fig
Yearly

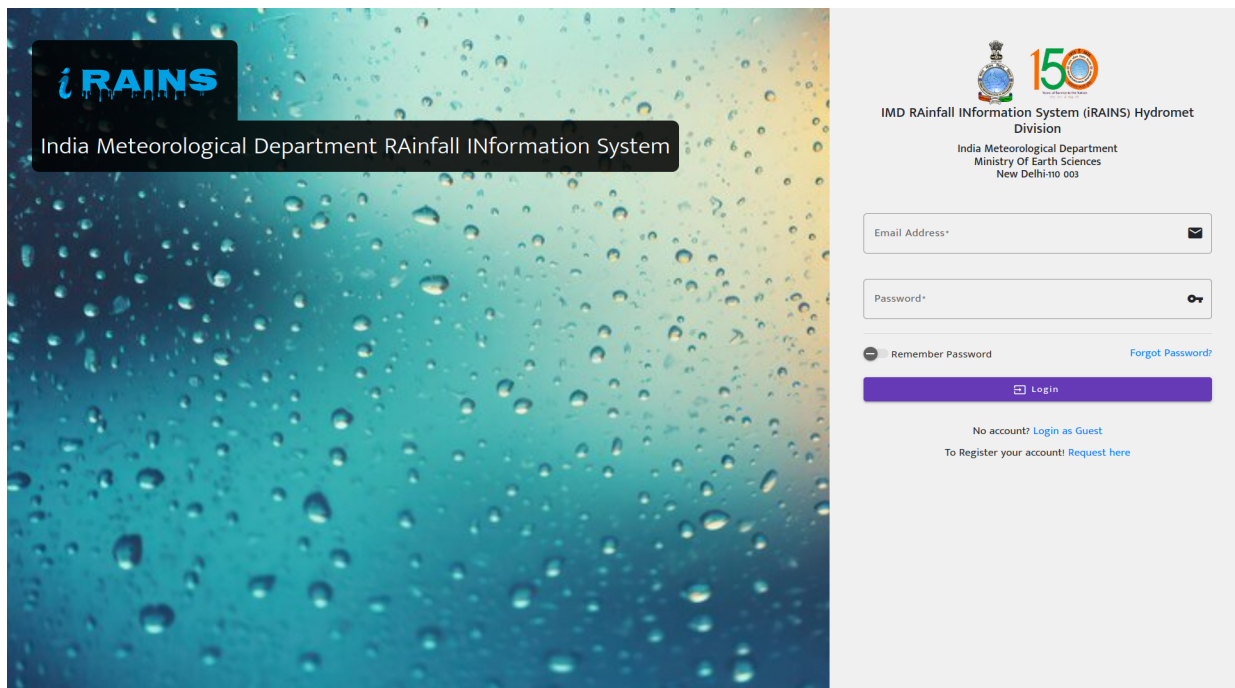


3.14 :

Station Statistics)

Login Page

- **Overview:** The Login Page serves as the gateway to IRAINS, designed to ensure secure and role-based access. Users are required to enter a username and password, with access permissions tailored to their roles. Additionally, a guest access option is available for users without credentials.
- **Access Levels:**
- **HQ Login:** Provides full access to all features, enabling HQ to manage, view, and disseminate rainfall data across all regions.
- **MC/RMC Login:** Offers access to features specific to MC/RMC roles, such as data entry and station-specific reporting.
- **Guest Access:** Grants limited viewing privileges to non-registered users.
- **New User Registration:** A dedicated registration link allows new IMD personnel to streamline their access to the system.



iRAINS
India Meteorological Department RAINfall INformation System

IMD Rainfall Information System (iRAINS) Hydromet Division
India Meteorological Department
Ministry Of Earth Sciences
New Delhi-110 003

Email Address*

Password*

☐ Remember Password [Forgot Password?](#)

[Login](#)

No account? [Login as Guest](#)
To Register your account! [Request here](#)

(fig 3.15 : Login Page)

4. Importance and Use

IRAINS empowers Hydromet Division by consolidating rainfall data into a single, accessible platform. Through its mapping capabilities, analytical tools, and comprehensive reporting features, IRAINS supports informed decision-making and real-time data management. It enables address challenges in hydrology and weather forecasting, making it a vital tool for tracking and responding to rainfall patterns, assessing seasonal variations, and preparing for extreme weather events.
